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A COMPREHENSIVE CFD INVESTIGATION OF GRID FINS AS EFFICIENT CONTROL SURFACE DEVICES

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ABSTRACT

Recently, grid fins have been receiving increasing attention as a practical and efficient means of controlling missile trajectory. Preliminary studies at IAR have demonstrated that modern CFD methods can be used for computing flows past complex grid fin type configurations, and that these methods can bring improved predictions over the earlier vortex lattice and/or shock expansion methods.

The current paper addresses the issue of the grid fin size, in terms of both the panel thickness and the frontal shape. The study covers three thicknesses for the grid fin panel, with front shapes having a simple blunt square face, as well as a sharp knife-edge shape. In addition, an important aspect of the present investigation is to quantify the aerodynamic effect of the ramp fairing installed immediately upstream of the blunt base upon which the grid fin assembly resides. A comparison of flow field characteristics and aerodynamic coefficients of the grid fin assembly, with and without the fairing ramp, would provide a direct means of evaluating the effect of the ramp. At this stage, the investigations are based on Euler calculations. The present study focuses on a standard grid configuration mounted on a generic cylindrical body.

NOMENCLATURE

C_A	Axial force coefficient
C_m	Pitching moment coefficient
C_N	Normal force coefficient
t	Web thickness of grid fin
δ	Deflection angle of grid fin
D	Diameter of the missile body
M_∞	Freestream Mach number

1 INTRODUCTION

For most missile designs, aerodynamic control surfaces are used to provide the required forces and moments to maneuver along the desired trajectory. Missile wings and fins must be able to provide accurate differential force and moment inputs as well as withstand extreme

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loads under high g turns. It has been shown earlier that grid fins do have the ability to generate relatively higher normal loads [1,2,3,4], and that their mesh type of framework is structurally more suitable, although it causes higher axial force, than conventional fins. Earlier IAR studies had concentrated on thin panels for the grid fin (zero thickness for the computational mesh) and did not address such practical issues as the thickness of the panels, the shape of the support base or the leading edge profile of the panels. Some computations did include the overall angle of attack effects and the deflection angle of the grid fins. It was also shown at that time that present computational methods were more accurate than the earlier vortex lattice techniques or the shock expansion methods [8, 9] which use certain approximations to the theory to include 3D configurations at finite angles of attack. Those methods will certainly not be suitable to tackle the current finite thickness panels of the grid fins or the blunt and pointed support base at a variety of deflection angles.

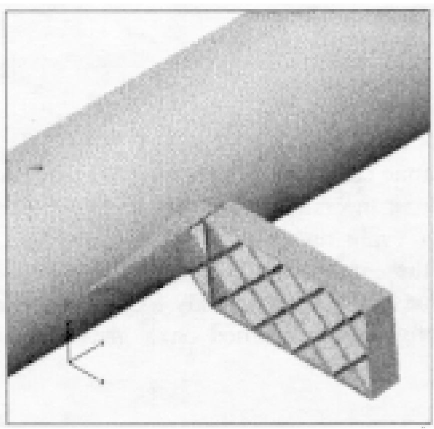

Figure 1. Schema of grid fin configuration with ramp

In order to understand some more practical issues of the grid fin concept, IAR and DREV have collaborated to comprehensively investigate the aerodynamic performance of a generic grid fin configuration, shown in Figure 1, which is similar to that used on the AA-12 missile. The chord and span of each panel is *0.175 D* as shown on Figure 1. The current investigation mainly addresses the following four issues:

- the effects from different thickness and frontal shapes of the grid panels
- the effects of the ramp fairing at the grid fin base
- the aerodynamic performance resulting from various control deflections of the grid fin
- the aerodynamic performance of the grid fin at different freestream Mach numbers.

Earlier grid fin studies [5,6,7], which investigated such design considerations, have been mainly experimental and have generally concentrated on external frame shape, web thickness, cross sectional profile and surface curvature parameters as the prime design variables. Reference 7 shows some of the web shapes and the grid fin configurations studied in these earlier experiments.

In Miller and Washington's [7] experimental study, for example, it was shown that frame shape had the greatest influence on normal force at subsonic Mach numbers whereas the web thickness affected the normal force at supersonic Mach numbers. It was also observed that noticeable drag reductions could be obtained through simple shaping of the web frontal profiles.

Obviously, an intense parametric design study, which could optimize grid fin frame cross section, web thickness and support structure, is somewhat expensive and impractical to be tackled experimentally. The computational studies along the lines of the present investigation may provide the efficient tool and the convenience for such performance envelope optimizations. It is envisaged that along with the superior lift, pitching and bending moment performance, a fully optimized grid fin and accompanying frame and support structure could provide drag levels typical of the conventional control fins.

Another feature investigated in the course of the present study is the introduction of a fairing immediately in front of the grid fin base. Since the grid fin assembly may need to be mounted on a thick block base attached to the missile surface, there will undoubtedly be some blockage and interference effects which emanate from the base and impact upon the grid fin performance. To suppress these unwanted aerodynamic effects, a ramp (Figure 1) was installed in the region immediately upstream of the grid fin assembly base. This fairing is similar to the ramp that is used on the Russian AA-12 missile with grid fin tail control.

The present study also investigates the effect of control deflection of the grid fin assembly (different rotational angles (δ) about the z-axis as shown in Figure 1). The aerodynamic performance for such configurations, when considered in the presence or in the absence of the fairing ramp in front of the grid fin base at different angles of attack, would be quite valuable in the overall judgement of grid fin performance in comparison to conventional control surfaces.

2 GRID FIN MESHES

ICEM CFD MULCAD was used to obtain the meshes on the grid fin assembly. The most challenging issue in the grid generation is the block construction to facilitate the finite thickness of each panel, and also, to suitably account for the front shape (block or wedge) of each panel. Appropriate attention in mesh communication must also be paid to an appropriate marriage between the fairing and the base block. Since the primary focus of this study was the actual grid fin assembly, the supporting missile body was modeled as a generic cylindrical body, thus easing some restraints on grid topology which would otherwise be present if the missile forebody had to be modeled accurately as well.

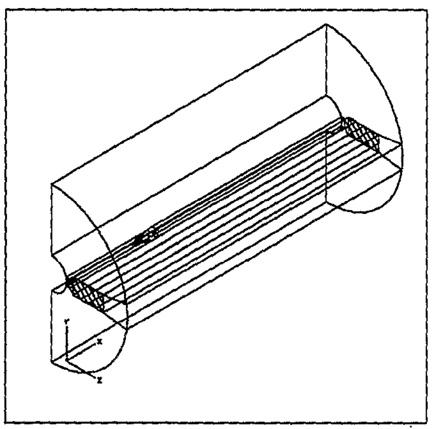
Figure 2. Mesh topology of grid fin

The grid fin topology as shown in Figure 2 essentially describes a symmetry plane missile configuration with a single grid fin assembly. The basic mesh topology was constructed with 32 blocks and about 2 million grid points. The meshes for grid fin at non-zero rotational angles, with respect to the main missile, could be produced with minimum adjustment to the basic topology.

3 RESULTS AND DISCUSSIONS

The NPARC code obtained from the NPARC Alliance [10,11] was used for most of the flow studies past the grid fin assembly shown in Figure 1. Owing to lack of in-depth experimental data for this configuration some comparison studies were also carried out using the NPARC Alliance supplied code WIND.

The computations were carried out for Mach numbers of *1.5* and *2.0*, with the grid fin deflection angle varying from 0^o to 10^o. Three panel thicknesses with respect to missile diameter *0.00375*, *0.00925* and *0.0209* were implemented on the various meshes for the standard grid fin configuration. The parent missile geometry, upon which the grid fin assembly is mounted, was configured as a simple cylinder facing freestream flow at zero degrees. The present computations are based on Euler equations as modeled in NPARC and/or WIND. The complete computational schedule is provided in Table 1.

Table 1 Computational schedule for grid fin

T/D	δ	*M*	Ramp	Fin Edge
0.00375	*5*	*2.0*	*Y*	*Flat*
0.00375	*5*	*1.5*	*N*	*Pointed*
0.00375	*5*	*2.0*	*N*	*Pointed*
0.00375	*5*	*1.5*	*Y*	*Pointed*
0.00375	*5*	*2.0*	*Y*	*Pointed*
0.00375	*10*	*2.0*	*Y*	*Pointed*
0.00925	*5*	*1.5*	*Y*	*Pointed*
0.00925	*5*	*2.0*	*Y*	*Pointed*
0.0209	*5*	*1.5*	*Y*	*Pointed*
0.0209	*5*	*2.0*	*Y*	*Pointed*

It should be noted that the axial force coefficient C_A, normal force coefficient C_N and pitching moment coefficient C_m used in the following text are all non-dimensionalised by following parameters:

Reference length: Cylinder diameter, D
Reference area: Cylinder cross section area
Reference point: *11.5D* ahead of the fin along the cylinder axis.

The axial force is along the x-axis (Figure 1) and the normal force is along the y-axis. The loads and moments were obtained by the integration of pressure distribution on a single grid fin assembly.

3.1 Effects of the ramp fairing

One of the features investigated here is the effect of the fairing installed immediately upstream of the grid fin support block. Figure 3 shows the computed flow field based on Mach number distribution for the simple blunt base configuration at freestream Mach number M_∞=*2.0*. The region in front of the blunt base supports a strong stagnated bow shock in addition to the presence of some oblique shocks on the grid fin surfaces.

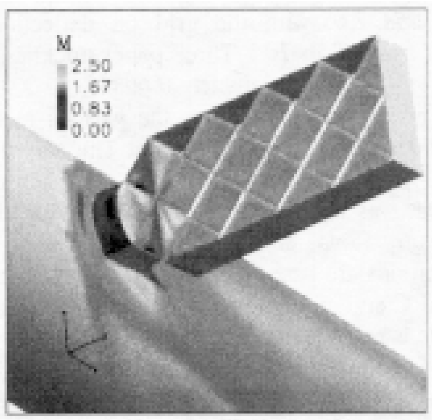

Figure 3. Mach number distribution of grid fin without ramp
M_∞=2.0, t/D=0.00375, δ=5°, Knife-edge

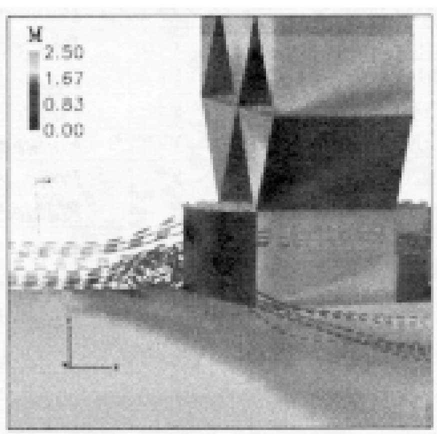

Figure 4. Flow pattern near leading edge of grid fin base without ramp
M_∞=2.0, t/D=0.00375, δ=5°, Knife-edge

Figure 4 gives the streamline pattern at the leading edge of the grid fin base. The appearance of a vortex immediately upstream of the forward facing blunt base is attributed to the dissipation terms present in the Euler equations. This vortex occupies almost half the height of the base. The vortex pattern near the trailing edge of the grid fin base is displayed in Figure 5. It shows a rather elongated vortex stretching in the streamwise direction with normal height of about three-quarter the size of the grid fin base. The presence of the bow shock and the two vortex system in front and aft of the base block, as well as the strong oblique shocks coming off from the grid fin panel corners have a profound effect upon the axial force performance of the missile.

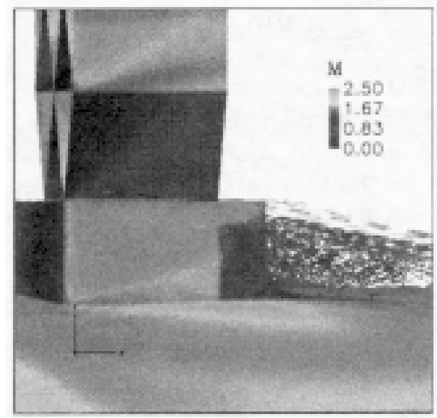

Figure 5. Flow pattern near backward edge of grid fin base
M_∞=2.0, t/D=0.00375, δ=5°, Knife-edge

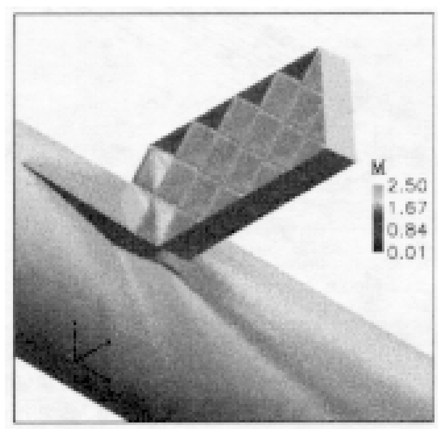

Figure 6. Mach number distribution of grid fin with ramp
M_∞=2.0, t/D=0.00375, δ=5°, Knife-edge

Figure 6, on the other hand, presents a much smoother flow field for the ramp-on configuration. There are no visible signs of strong shock waves interfering with the flow upstream of the ramp. Also the shock waves on the grid fin surfaces near the base are less affected by the near-body oncoming flow.

The axial and normal force coefficients as obtained from the pressure integration along the surfaces of the grid fin and its base are shown in Figures 7 and 8. It must however be understood that the presence of vortical flow in front of the forward facing blunt base in an inviscid computation may introduce certain inaccuracies when integrating the pressure loads. It is observed that the bottom-mounted blunt base contributes significantly to the axial force increase. In other words, the fairing installed in front of the base obviously has the effect of reducing the overall axial force and somehow increasing the normal force performance slightly.

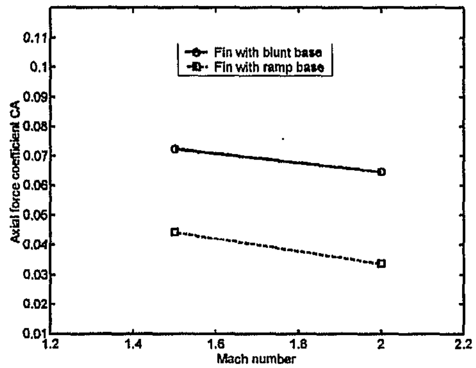

Figure 7. Comparison of axial force coefficient for fins with blunt base and ramp fairing
t/D=0.00375, δ=5°, Knife-edge

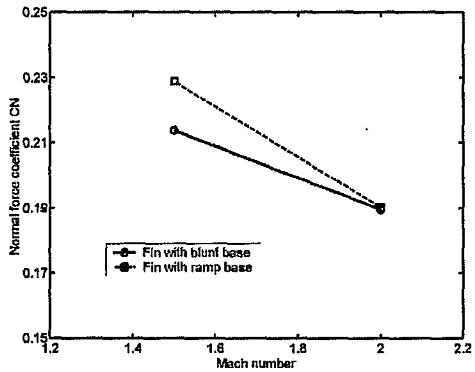

Figure 8. Comparison of normal force coefficient for fins with blunt base and ramp fairing
t/D=0.00375, δ=5°, Knife-edge

3.2 Effects of the web thickness

Web thickness has direct implications upon the structural integrity of the grid fin itself. An increase in web thickness may lead to a higher normal and shear stress threshold but might also adversely impact upon axial force and other aerodynamic characteristics of the grid fin. The present study seeks to quantify various aerodynamic effects in terms of the resultant aerodynamic loads and also to examine the detailed flow patterns in presence of these different web geometries. Towards this end, three web thicknesses: $t/D=0.00375$, 0.00925 and 0.0209 were selected for this investigation. The aerodynamic performance of the grid fin was examined for a fin deflection angle of $\delta=5^o$ with the base fairing installed.

Figure 9 shows the Mach number contours on a plane, which runs parallel to the missile axis and cuts centrally through the middle of the grid fin with panel thickness of 0.00375. The freestream Mach number in this case was $M_\infty=2.0$ on a configuration with a fairing and a fin deflection angle set at $\delta=5^o$. The fairing seems to guide the flow smoothly into the grid fin. The front and the back faces of the grid fin show a complex system of oblique shocks but most of the upper channels allow the flow to pass through at supersonic Mach numbers. Lower channels see the flow at relatively higher angles as it works its way up the ramp and there is some deceleration in the lower channels with the customary backward facing step flow in the aft base region.

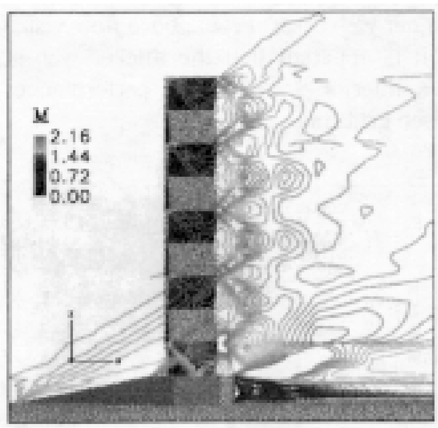

Figure 9. Mach number contours cutting through the fin centerline
M_∞=2.0, t/D=0.00375, δ=5°, Knife-edge

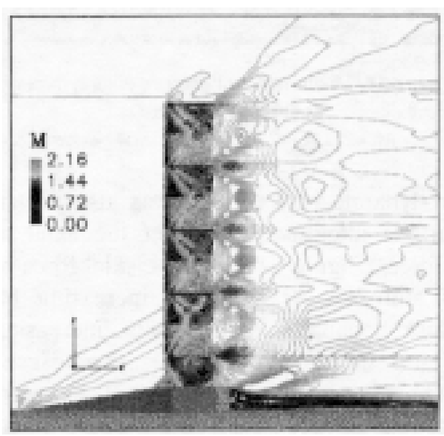

Figure 10. Mach number contours cutting through the fin centerline
M_∞=2.0, t/D=0.00925, δ=5°, Knife-edge

Figure 10 shows the same flow for a similar grid fin with a panel thickness of 0.00925. All other freestream conditions corresponded to the case described in Figure 9. The shocks at the front and backward face of the grid fin are more obtuse. There are more regions of increasingly decelerated flow with the blunt trailing edge of the base showing stagnated flow. It appears that in terms of the axial and normal force, this configuration would not be as preferable as the first one. Figure 11 captures the flow features for a grid fin with the thickest web width of *0.0209*. It appears that the flow is subsonic and virtually choked through most of the grid fin channels. Note also the presence of a standing bow shock in front of the grid fin assembly as it curves up and around the top of the grid fin. The flow as it emerges through most of the grid fin channels remains subsonic for some distance downstream of the grid fin. The retarded flow region aft of the blunt base seems to grow in comparison to other thinner web cases. From above flow visualization analysis it is apparent that the thicker web grid fins will have inferior aerodynamic performance to the thinner web grid fins.

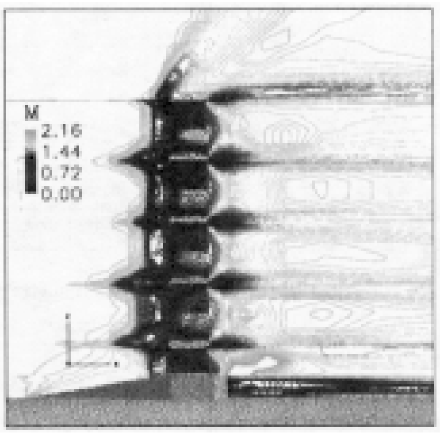

Figure 11. Mach number contours cutting through the fin centerline
M_∞=2.0,t/D=0.0209, δ=5°, Knife-edge

The aerodynamic performance was also measured in terms of the integrated values of the axial and the normal forces. Figure 12 shows the axial force plots for $M_\infty=1.5$ and 2.0. As expected, increasing blockage from thicker panels coupled with increasing flow interference and regions of decelerations cause higher axial force at larger web thicknesses. The presence of a large bow shock encompassing the entire geometry of the grid fin also lead to increasing axial force. The axial force at $M_\infty=1.5$ is consistently higher than that at $M_\infty=2.0$ which is consistent with the transonic similarity theory that axial force tends to peak around sonic flow.

The normal force behavior (Figure 13) is somewhat consistent with the axial force results. A flow field on similar geometry containing larger regions of decelerated and stagnated flow is unlikely to produce as much lift as the one showing smoother flow. The normal force diminishes with increasing web thickness.

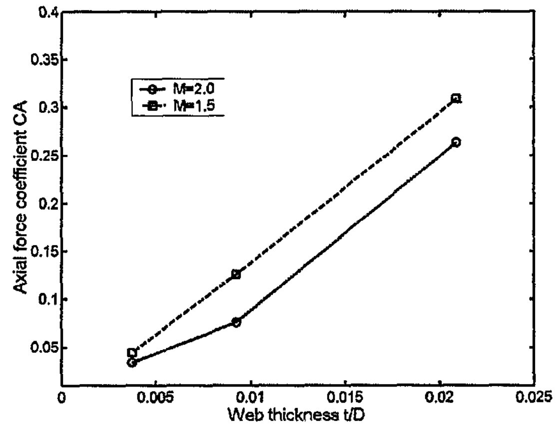

Figure 12 Axial force coefficients
M_∞=2.0, δ=5°, Knife-edge

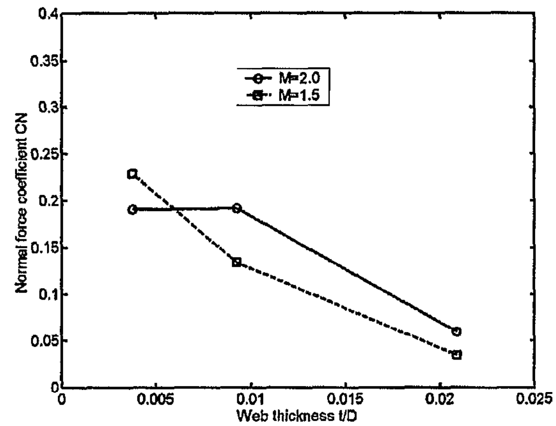

Figure 13 Normal force coefficients
M_∞=2.0, δ=5°, Knife-edge

Figure 14 presents the surface Mach number distribution on a grid fin with the web thickness of $t/D=0.0209$ at $M_\infty=2.0$. On the side surface of the grid fin assembly, the flow does not seem to be as uniform as the grid fins with thinner web panels, see for example the side flow past the grid fin with web thickness of *0.00375* in Figure 6. Strong oblique shocks emanating from stations corresponding to the cross points of fin webs are formed on the fin side surface. Another notable feature is the character of the bow shock on the missile surface itself. For the grid fin panel of this thickness, the strong bow shock appears to have widened on the missile body surface.

For the lower Mach number case of $M_\infty=1.5$, the oblique shock wave pattern on the side surface of the grid fin assembly begins earlier at a lower web thickness of $t/D=0.00925$ (Figure 15). However the shock intensity is not as strong as the case discussed in Figure 14. The bow shock on the missile surface however is just as strong as that seen in Figure 14, which means that choking effects are also present for smaller panel thicknesses but at lower Mach numbers.

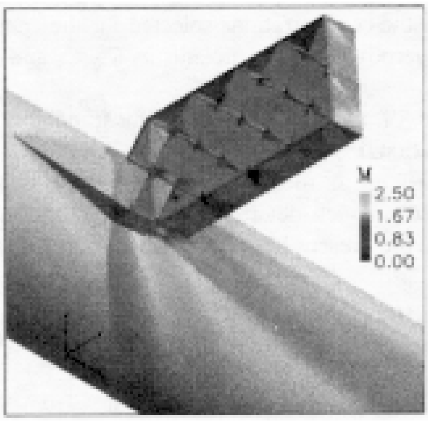

Figure 14 Surface Mach number distribution for the web thickness $t/D=0.0209$ $M_\infty=2.0$, $\delta=5^o$, Knife-edge

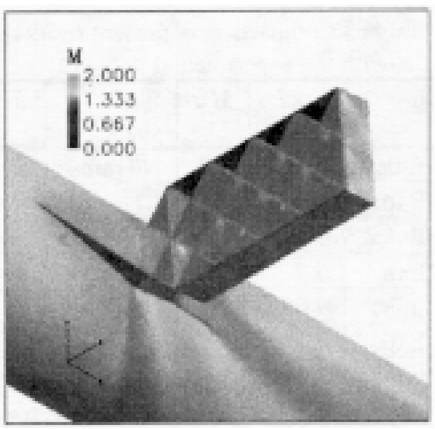

Figure 15 Surface Mach number distribution for the web thickness $t/D=0.00925$ $M_\infty=1.5$, $\delta=5^o$, Knife-edge

3.3 Effects of the fin deflection angle

This section investigates the performance of the grid fin as an effective control surface device. Its performance is studied at a number of deflection angle (δ) settings with reference to the missile angle of incidence. The missile was always fixed at zero degree angle of attack.

Computations were carried out for three deflection angle settings; $\delta=0^o$, 5^o and 10^o for a grid fin assembly with panel thickness of $t/D=0.00375$ and free stream Mach number of $M_\infty=2.0$. For the base line computation at $\delta=0^o$, the grid fin panels were equipped with a blunt leading edge. For the grid fin configurations at higher deflection angles the panel leading edges were given a knife-edge pointed profile. So the results at $\delta=5^o$ and 10^o degrees contained the composite effects from the deflection and the leading edge profile.

Table 2 Integration results for different deflection angles

δ	C_A	C_N	C_m
0^o	0.0185	0.0	-0.00202
5^o	0.0337	0.1902	-2.1876
10^o	0.0866	0.3981	-4.6287

Table 2 lists the axial force, normal force and pitching moment coefficients at different deflection angles. The integration data show that with the increase of the deflection angle δ from 0^o to 10^o, the normal force coefficient of the fin increases from 0.0 to 0.3981. Correspondingly there is an almost 5 fold jump in axial force for the same range of grid fin deployment. The comparison is rather qualitative because the panel geometry at leading edge is different.

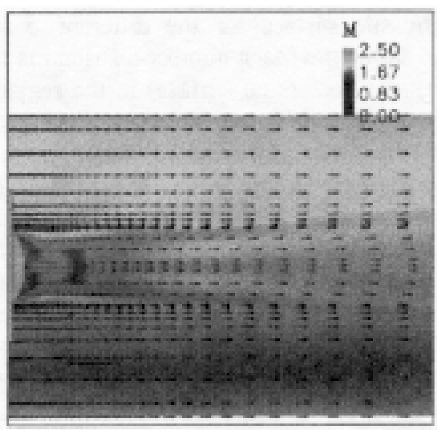

Figure 16 Mach number and velocity vector distribution on missile surface at $\delta=0^o$ $M_\infty=2.0$, $t/D=0.00375$, Flat-edge

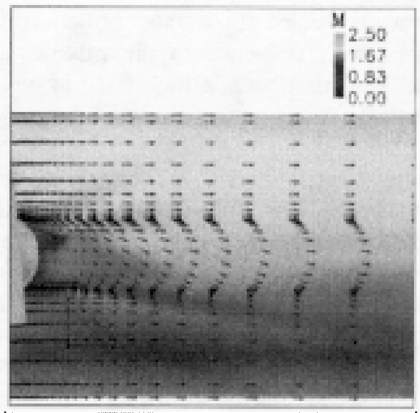

Figure 17 Mach number and velocity vector distribution on the missile surface at δ=5°
M_∞=2.0, t/D=0.00375, Knife-edge

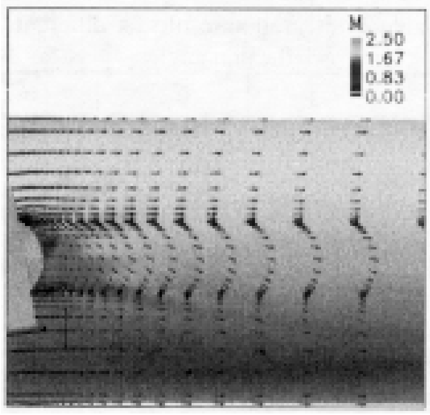

Figure 18 Mach number and velocity vector distribution on the missile surface at δ=10°
M_∞=2.0, t/D=0.00375, Knife-edge

The effect of the fin deflection angle was further scrutinized by looking at the Mach number distribution on the missile surface for the different δ settings. Figure 16 shows the Mach number distribution (as well as the velocity vectors on surface) in the region aft of the grid fin assembly for freestream Mach number $M_\infty=2.0$ and $\delta=0^o$. As expected the flow is fairly symmetrical about the centerline, and there are distinct regions of stagnated and vortical flow behind the grid fin base. It is apparent from a similar flow field snap shot for $\delta=5^o$ setting as seen in Figure 17, that the flow is increasingly more asymmetric with the decelerated flow being pushed more towards one side of the base. The same phenomenon seems to continue as the deflection angle is increased to $\delta=10^o$ (see Figure 18) as the wake also gets pushed to one side. The extent of the stagnated flow aft of the base also seems to decrease with increasing deflection angle. The asymmetry obtained here of course can be corrected somewhat by the traditional four-fin deployment of the grid fins. Pairing fins displaced in opposing clockwise sense will correct this anomaly, whereas synchronous movement in same direction will aggravate this problem. On actual tail-control missiles with grid fins, the missile base is not too distant from the grid fins, and the aforementioned flow complexities aft of the grid fin assembly will also have profound implications on the base axial force.

3.4 Comparison with experimental data

The experimental data of Simpson and Sadler [12] was chosen for some of the present comparison. Table 3 itemizes the configurations selected for the comparison of such aerodynamic coefficients as C_N, C_A and C_m.

The use of a ramp fairing in most of the present computational model renders this comparison even more difficult as the configurations in reference [12] contained no such device. One possible option was to compare the present blunt base, ramp-less case at:

(i) $\alpha = 0^o$ and $\delta = 5^o$ or
(ii) $\alpha = 0^o$ and $\delta = 10^o$ both cases at freestream $M_\infty=2.0$,

against the following experimental data from Simpson and Sadler [12]:

(i) $\alpha = 5^o$ and $\delta = 0^o$ or
(ii) $\alpha = 10^o$ and $\delta = 0^o$ at $M_\infty=1.8$ and 3.5

Table 3 Comparison of present results with the experimental data [12]

		Coef.	M_∞	Fin shape	δ or α
C_N	Euler	0.190	2.0	No ramp	$\delta=5^0$
	Exp.	0.15	3.5	L1	$\alpha=5^0$
	Euler	0.398	2.0	No ramp	$\delta=10^0$
	Exp.	0.3	1.8	L3	$\delta=10^0$
C_A	Euler	0.065	2.0	No ramp	$\delta=5^0$
	Exp.	0.09	1.8	L3	$\alpha=5^0$
C_m	Euler	2.181	2.0	No ramp	$\delta=5^0$
	Euler	2.188	2.0	Ramp	$\delta=5^0$
	Exp.	1.5	3.5	L1	$\alpha=5^0$
	Euler	4.629	2.0	Ramp	$\delta=10^0$
	Exp.	3.0	3.5	L1	$\alpha=10^0$

The measured normal forces as shown in Table 3 were selected from Figure 4 of Reference 12, and actually refer to the load on a single fin. The axial force and pitching moment in Reference 12, however, had to be appropriately scaled down, because the axial force refers to the resultant force on all four-grid fins installed on a complete missile and the pitching moment contains the total moment from two horizontal fins.

The comparison between measured and computed normal force appears to be reasonable. The computed normal force coefficient was generally a little higher than the measured data, whereas, the axial force was lower than the measured data. There are several reasons, which may explain the higher values of the computed normal force coefficient. The knife edge type profile of the panels used in current calculations is more aerodynamic and tends to produce smoother flow leading to a higher normal force than what would be available from the blunt leading edge shaped panels. This would have similar implications for the axial force, in that, knife-edged delta face will naturally produce less axial force than a block blunt shape. In addition, when non-dimensionalised by the missile body cross section area, the fin size in the computational model is somewhat larger than the one used in experiment, which too might be responsible for some additional normal force. Other difference can be attributed to the Mach number differences and other 3D and asymmetry effects presented in the complete model.

In terms of the axial force comparisons, it is also worth noticing that the height of the current grid fin base is somewhat bigger than the experimental geometry, which is supposed to increase the axial force of the grid fin assembly. But the fairing in front of the grid fin base, for present model, contributed a lot to reduce the axial force. Together with the knife leading edged grid fin panel geometry, the overall axial force performance of present grid fin model is better than the experimental model.

The pitching moment coefficient obtained from the current geometry too seems to be larger than the corresponding values reported in the experiment. This may be explained in terms of the improved aerodynamics through the grid fin channels used in present models, which as observed above produce relatively larger normal loads leading to higher pitching moments.

4 CONCLUSIONS

CFD computations have been carried out to study the aerodynamic effects of a fairing ramp installed at the grid fin base. The investigation also included the aerodynamic effects from various thicknesses of the grid fin panels, as well as the different rotational deployments of grid fin. Following conclusions have been reached:

- The fairing installed in front of the base of the grid fin assembly can reduce the blockage and other interference effects. This results in some improvement in aerodynamic performance, including some increase in lift and reduction in axial force. There are distinct regions of stagnated vortical flow, which in absence of the fairing, are responsible for the degradation of the aerodynamic performance. Further Navier-Stokes computations would be required for a more comprehensive performance studies of the present grid fin configurations.

- Increase of grid fin panel thickness, produces bow shocks in front of the grid fins and on missile surfaces coupled with strong flow interference inside the channels. This leads to degradation in grid fin performance.

- Grid fin rotation has a noticeable affect upon the character of the wake aft of the grid fin base. There are other asymmetric changes to the nature and extent of the vortical flow behind the blunt base. This also introduces some axial force penalties and control surface challenges, some of which would have to be treated appropriately with four fin deployment.

- It was noticed that for thin panel geometry ($t/D=0.00375$), both the normal and axial forces decreased with an increase in Mach number from $M_\infty=1.5$ to 2.0, which in a global sense seems to be consistent with the rules of the transonic theory. For larger thickness panels which support strong bow shocks and other interference effects, the trend for normal force was reverse.

- The absence of relevant experimental data near current flow conditions, makes the validation exercise somewhat difficult, however, qualitative comparison against available measurements found in Reference 12 are quite encouraging.

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REFERENCES

1 Sun, Y. and Khalid, M. (1997), "Prediction of Supersonic Flow around Grid Fin Missile Using NPARC", The Fifth Annual Conference of the Computational Fluid Dynamics Society of Canada, CFD 97 May 25-27.

2 Sun, Y. and Khalid, M. (1997), "A CFD investigation of Grid Fin Missiles", AIAA 98-3568, 34th, AIAA/ASME/SAER/ASEE Joint propulsion Conference & Exhibits, July 13-15, Cleveland, OH.

3 Khalid, M., Sun, Y. and Xu, H. (1998), "Computation of Flows Past Grid Fin Missiles", Symposium of the AVT on "Missile Aerodynamics", Sorrento, Italy, May 11-15.

4 Xu, H., Khalid, M., Chen, S. and Lesage, F. (1999), "Investigation of Grid Fins as Modern Control Surface Devices for Enhancing Missile Aerodynamic performance", The 46th Annual Conference of Canadian Aeronautics and Space Institute, May 3-5, Montreal, Quebec, Canada

5. Washington, D. W., Booth, P. F. and Miler, M. S. (1993), "Curvature and leading edge sweep back effects on grid fin aerodynamic characteristics", AIAA-93-3480-CP.

6. Washington, D. W., and Miller M. S. (1993), "Grid Fins - A New Concept for Missile Stability and Control", 31st Aerospace Sciences Meeting & Exhibits', AIAA 93-0035, January 11-14, Reno, NV.

7. Miller, M. S. and Washington, D. W. (1994), "An experimental investigation of grid fin drag reduction techniques", AIAA-94-1914-CP.

8. Brooks, R. A. and Burkhaltar, J. E. (1989), "Experimental and Analytical Analysis of Grid Fin Configurations," Journal of Aircraft, Engineering Note, September.

9. Burkhaltar, J. E. (1996), "Grid Fins for Missile Applications in Supersonic Flow", 34th Aerospace Sciences Meeting & Exhibits, January 15-18, Reno, NV.

10. NASA LeRC/AEDC Alliance (1996), "NPARC User's Guide", Version 3.0, Sept., Washington.

11. Cooper, G. K. and Sirbaugh, J. R. PARC Code: Theory and Usage AEDC-TR-89-15, Sverdrup Technology Inc.

12 Simpson, G. M. and Sadler, A. J. (1998), "Lattice controls: A comparison with conventional, planar fins", RTO-MP-5, Missile Aerodynamics, Sorrento, Italy